

Chapter 9

Non-Linear Pricing, Natural Monopoly, and Game Theory

"In business, the rearview mirror is always clearer than the windshield."

— Warren Buffett

Before You Begin: Quick Self-Assessment

Answer each question before reading. If you cannot, review the indicated earlier section.

1. **What is a monopolist's profit-maximising condition?** *Answer:* Set $MR = MC$ and read off the demand curve for the profit-maximising price. (Ch. 8)
2. **Define consumer surplus geometrically.** *Answer:* The area between the demand curve and the price line, up to the quantity purchased — the triangle above the price and below the demand curve. (Ch. 4)
3. **Under uniform monopoly pricing, does the market achieve allocative efficiency?** *Answer:* No. The monopolist charges $P > MC$, restricting output below the competitive level and creating dead-weight loss. (Ch. 8)
4. **What is a fixed cost?** *Answer:* A cost that does not vary with output in the short run — incurred even if the firm produces zero units. (Ch. 6)
5. **If two players both have a "dominant strategy," can they still be trapped in a mutually bad outcome?** *Answer:* Yes — the prisoner's dilemma is the classic example. Playing your dominant strategy regardless of what the other player does can lead to a Nash equilibrium that is Pareto dominated by the cooperative outcome. (Preview of Ch. 9)

Questions This Chapter Will Answer

- Why does Amazon Prime charge an annual fee *plus* a usage charge — and why does this structure double monopoly profit?
- Electricity grids have declining average costs throughout. Why can't they survive if forced to price at marginal cost?
- How can a railway charge business travellers more to subsidise leisure fares — and is this fair?
- Two petrol stations both want to charge high prices. Why do they end up in a price war anyway?
- What makes (Confess, Confess) a Nash equilibrium when both prisoners would prefer (Silent, Silent)?
- When does individual rationality produce collectively irrational outcomes?

Chapter Road Map. This chapter extends the monopoly pricing analysis from Chapter 8 in three directions. First, **non-linear pricing** shows how a firm with market power can increase profit by splitting its pricing into a fixed access fee and a per-unit charge — two-part pricing doubles monopoly profit while eliminating deadweight loss. Second, **natural monopoly** examines industries where average cost falls across the entire market output range, creating a fundamental tension between efficiency ($P = MC$ causes losses) and financial viability ($P = AC$ is sustainable but slightly inefficient). Third, **game theory** provides the tools to analyse markets with two or more strategic firms: Nash equilibrium, dominant strategies, the prisoner's dilemma (individual rationality produces collective irrationality), and coordination games (multiple equilibria, coordination failure risk).

Learning Objectives and Topics

Each objective states what you will be able to *do*.

LO1 Distinguish linear from non-linear pricing; identify fixed fee and usage charge in real examples.

§9.2 *Non-Linear Pricing*

LO2 Apply the optimal two-part pricing rule: set $p^* = MC$, $F^* = CS(p = MC)$; prove profit doubles relative to uniform monopoly.

§9.3 *Two-Part Pricing*

LO3 Explain why natural monopolies face a regulatory dilemma; compare unregulated, MC-pricing, AC-pricing, and two-part outcomes.

§9.4 *Natural Monopolies*

LO4 Explain how cross-subsidisation allows a regulated natural monopoly to break even without a government subsidy.

§9.4 *Natural Monopolies*

LO5 Identify players, strategies, and payoffs; solve for Nash equilibria and dominant strategies using the best-response / circle method.

§9.5 *Game Theory*

LO6 Explain why the prisoner's dilemma produces a collectively suboptimal Nash equilibrium; apply to pricing, advertising, and arms races.

§9.5 *Game Theory*

Notation Used in This Chapter.

$T(q)$	total payment for q units	F	fixed (entry) fee
p	per-unit usage charge	p^*, F^*	optimal two-part values
$CS(p)$	consumer surplus at price p	Q_A	unregulated monopoly qty
Q_R	regulated ($P = MC$) quantity	Q_{AC}	break-even quantity
FC	fixed cost	(s_A, s_B)	strategy profile
$\pi_i(s_A, s_B)$	player i 's payoff	s^*	Nash equilibrium strategy

9.1. Introduction

Week 8 showed that a monopolist earns profit \$10 with uniform pricing and destroys \$5 in deadweight loss. This week asks: without knowing each customer's exact willingness to pay, can the firm do better?

The Pricing Toolkit: From Week 8 to Week 9

Strategy	Info needed	Profit	CS	DWL
Uniform monopoly	None	π_M	> 0	> 0
Market separation	Observe group	$> \pi_M$	> 0	reduced
Two-part pricing	Aggregate demand	$2\pi_M$	0	0
Perfect price discrimination	Individual WTP	$CS_{PC} + \pi_M$	0	0

Two-part pricing is the week's centrepiece: it doubles profit using only aggregate demand information — no need to observe individual types.

This chapter extends the analysis with two structurally new topics. *Natural monopoly* asks what happens when the efficient market structure is a single firm due to cost conditions — the regulatory implications are profound. *Game theory* asks what happens when there are multiple strategic firms — individual optimisation is no longer sufficient.

We start by distinguishing linear from non-linear pricing.

9.2. Non-Linear Pricing

A coffee shop charges \$1.60 for an 8 oz. cup (\$0.20/oz.) and \$2.00 for a 16 oz. cup (\$0.125/oz.). The larger cup is not twice the price. Why does this happen, and why does it help the seller?

9.2.1. Linear vs. Non-Linear Pricing

Linear Pricing: $T(q) = p \cdot q$ — total payment is exactly proportional to quantity. Per-unit price is constant.

Non-Linear Pricing: $T(q)/q$ is not constant — the per-unit price changes with quantity. The pricing schedule tracks the shape of the demand curve more closely than any single flat price can.

In words: Non-linear pricing charges high prices for early units (where WTP is highest) and lower prices for later units. It extracts more total surplus than a flat price.

Examples: Costco bulk packs, tiered utility rates, size-based coffee pricing, quantity discounts.

9.2.2. Two-Part (Two-Tier) Pricing

The intuition before the math. Under uniform monopoly pricing, the firm charges a single price and produces the quantity where $MR = MC$. This generates a profit rectangle — but it also leaves a large triangle of consumer surplus uncaptured. Consumers who value the good highly pay the same price as those who barely find it worth buying, and that surplus stays with consumers rather than flowing to the firm.

Two-part pricing is a way to capture that triangle without perfect information about individual willingness to pay. The insight is to split the pricing decision into two separate levers: (1) set the *per-unit charge* p to equal marginal cost — this maximises the total quantity sold and hence the total surplus available in the market; (2) charge a *fixed access fee* F equal to the entire consumer surplus generated at $p = MC$ — this transfers that surplus to the firm via a lump-sum payment. The consumer still buys (they are indifferent: their net gain is exactly zero), but the firm captures everything.

Think of it this way. A gym sets a \$0 fee per visit and charges a monthly membership of \$60. The zero per-visit price means members work out as much as they like — efficient from society's perspective. The membership fee skims off the value they placed on that access. The gym earns more than it ever could by charging per visit.

Two-Part Pricing: The consumer pays:

- a **fixed fee** F to gain access (independent of quantity); and
- a **per-unit charge** p for each unit purchased.

$$T(q) = F + p \cdot q \quad (9.1)$$

In words: The fixed fee extracts surplus; the per-unit charge sets how much is consumed. Setting p low (= MC) maximises quantity — and therefore the surplus available to extract as F .

Amazon Prime. Annual fee $\approx$ \$139 (the fixed fee F) to access free two-day shipping, Prime Video, and discounts. Per-qualifying-purchase shipping charge = \$0 (the usage charge p). Amazon sets F high enough to extract frequent shoppers' surplus; the zero per-unit charge encourages maximum usage — exactly the logic of optimal two-part pricing. Walmart+, Costco memberships, and cloud computing platforms use the identical structure.

Check Your Understanding

1. A gym charges \$50/month plus \$5/class. Is this linear or non-linear? What is F ? What is p ?
2. 500 g flour costs \$2.00; 1 kg costs \$3.60. Calculate per-gram price for each. Is this linear pricing?

3. True or false: “A lower per-unit charge always means higher profit for the firm under two-part pricing.”

Two-part pricing’s power comes from separating the access decision (fixed fee) from the quantity decision (usage charge). The worked example shows exactly how this doubles monopoly profit.

9.3. How Two-Part Pricing Works: A Worked Example

A monopolist earns \$10 with uniform pricing. The efficient output would generate \$20 in consumer surplus. Can the monopolist capture that entire \$20 without perfect information about individual willingness to pay?

9.3.1. Benchmark: Three Scenarios Compared

Inverse demand: $P = 3 - Q/10$. Constant $MC = AC = \$1$.

Standard monopoly: $MR = 3 - Q/5 = 1 \Rightarrow Q_M^* = 10, P_M^* = \$2, \pi_M = \$10$.

Perfect competition: $P = MC \Rightarrow Q_{PC} = 20, CS_{PC} = \frac{1}{2}(3 - 1)(20) = \20 .

Two-part pricing: set $p^* = MC = \$1$; output = $Q_{PC} = 20$; $F^* = CS(p = MC) = \$20$; $\pi_{2P} = \$20$.

Optimal Two-Part Pricing Rule:

1. Set $p^* = MC$ (achieves efficient output; maximises the surplus pie).
2. Set $F^* = CS(p = MC)$ (extracts the entire pie through the fixed fee).

Total profit = $F^* = CS(p = MC)$. **Two-part profit is always exactly twice uniform monopoly profit for linear demand.**

In words: Step 1 makes the pie as large as possible. Step 2 takes the whole pie as the entry fee.

“Setting $p = 0$ maximises two-part profit.” It is tempting to think lower p means higher F . But $p < MC$ forces the firm to lose money on every unit sold. The loss on units sold more than offsets the larger fixed fee. Optimal is $p^* = MC$, not zero.

$\pi_{2P} = 2\pi_M$ for linear demand: the monopoly profit rectangle is exactly half the CS triangle. Two-part pricing captures both halves.

9.3.2. The Three-Scenario Comparison Table

“Which Pricing Regime Am I In?” — The Week 9 Decision Guide

Scenario	Q	P	CS	π	TS	DWL
Standard monopoly	10	\$2	\$5	\$10	\$15	\$5
Perfect competition	20	\$1	\$20	\$0	\$20	\$0
Two-part pricing	20	\$1+fee	\$0	\$20	\$20	\$0

Two-part pricing matches perfect competition on efficiency (TS = \$20, DWL = 0) but matches PPD on equity (CS = 0, monopolist takes everything). The pie is as large as possible; the firm eats the whole pie.

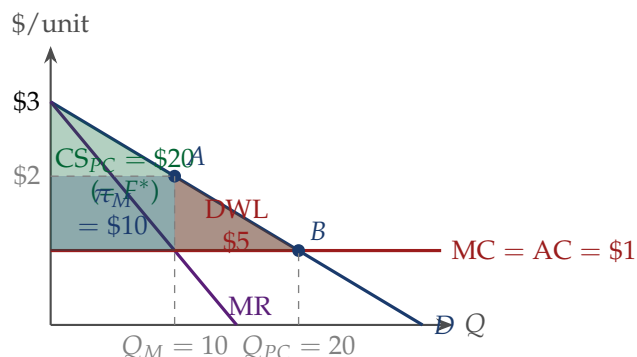


Figure 9.1. Two-part pricing benchmark: $P = 3 - Q/10$, $MC = \$1$. What to look for: Standard monopoly profit (blue rectangle, \$10) is exactly half the CS triangle (green, \$20). Two-part pricing captures the *entire* green triangle as the fixed fee $F^* = \$20$, eliminating the DWL triangle (red, \$5). The firm doubles its profit; consumers get nothing.

In plain English:
Two-part = efficient like competition, inequitable like PPD. The monopolist doubles profit without needing to know individual WTP.

9.3.3. Heterogeneous Consumers: A Worked Example

The clean result above assumes all consumers are identical. Real markets have consumers with different levels of demand. The firm must then choose between (i) setting a *type-specific* fee (if types are observable), (ii) setting a *uniform* fixed fee that retains both types, or (iii) *excluding* low-demand consumers and charging only high-demand ones. The trade-off between these options is central to real pricing strategy.

Gym Membership: Two Consumer Types.

A town has one gym. Half the residents are **Fitness Nuts** (Type H) with individual inverse demand $P_H = 20 - Q_H$. Half are **Casual Exercisers** (Type L) with individual inverse demand $P_L = 12 - Q_L$. The gym's $MC = AC = \$2$ per visit. Types are not directly observable.

Step 1: Consumer surplus at $p = MC = \$2$.

- Type H: $Q_H = 20 - 2 = 18$ visits; $CS_H = \frac{1}{2}(20 - 2)(18) = \162 .
- Type L: $Q_L = 12 - 2 = 10$ visits; $CS_L = \frac{1}{2}(12 - 2)(10) = \50 .

Option A — Type-specific fees (types are observable): Set $F_H = CS_H = \$162$ for Fitness Nuts; $F_L = CS_L = \$50$ for Casual Exercisers. Both types join; per-pair profit = $162 + 50 = \$212$.

Option B — Uniform fee, retain both types: Set $F = CS_L = \$50$ (the fee that keeps the low type from being excluded). Fitness Nuts pay \$50 even though their CS is \$162 — they enjoy *information rent* of \$112. Per-pair profit = $2 \times 50 = \$100$.

Option C — Exclude low types: Set $F = CS_H = \$162$; only Fitness Nuts join. Per-pair profit = $1 \times 162 = \$81$ (half the consumers excluded).

Ranking: Option A (\$212) > Option B (\$100) > Option C (\$81). Observing types is most profitable. When types are unobservable, the uniform-fee strategy retains both but surrenders rent to the high type.

Business insight: This is why firms invest heavily in loyalty programmes, usage data, and tiered membership structures — any information that helps identify high-demand users allows a

shift toward Option A, dramatically increasing profit.

With heterogeneous consumers, the firm often combines two-part pricing with cross-group subsidisation: electricity companies charge residential customers one fixed connection fee and commercial customers another. Profits from commercial customers cross-subsidise residential service — a practice examined in depth in the natural monopoly section below.

Why This Matters. Two-part pricing is one of the most widespread pricing architectures in modern business: gym memberships, streaming services, cloud computing, ride-sharing programmes, software subscriptions. Understanding the logic helps consumers assess whether a membership is genuinely worthwhile (is your $CS > F$?), and helps regulators evaluate whether access fees exclude lower-income consumers from essential services.

Application: Two-Part Pricing Across Industries.

Two-part pricing is not a niche strategy — it is the dominant architecture in many of the largest industries in the modern economy. The table below shows how the two-part structure appears across very different business contexts.

Company / Product	Fixed Fee F	Per-Unit Charge p
Amazon Prime	\$139/year membership	\$0 shipping on eligible items
Costco / Sam's Club	\$65/year membership	In-store prices (low mark-up)
SaaS platforms (Salesforce)	Monthly platform fee	Per-user or per-transaction fee
Cloud computing (AWS)	Free tier / account fee	Per GB, per CPU-hour
Golf / tennis clubs	Annual membership	Green fee or court fee per use
Amusement parks (Disney+)	Annual pass / subscription	In-park food and merchandise

In every case the firm sets the usage charge at or near marginal cost to maximise participation and total surplus generated — then captures that surplus through the fixed fee. For business managers, this model is powerful because it converts variable-fee revenue (which fluctuates with usage) into more predictable membership revenue, improving revenue forecasting and customer lock-in.

Management implication: When considering a two-part pricing structure, the key strategic question is: “How large is the average consumer’s surplus at $p = MC$?” That is the maximum F the market will bear. If the answer is “very large” (as with Amazon Prime shoppers), the membership model is highly profitable. If it is small or heterogeneous, the firm must be careful not to exclude lower-demand consumers.

Real-world tiered memberships (Gold, Silver, Basic) are attempts to approach Option A pricing when types are not directly observable. Each tier is designed to self-select a particular consumer type.

Check Your Understanding

1. Using the benchmark ($P = 3 - Q/10$, $MC = \$1$): if the firm mistakenly sets $p = \$1.50$, compute demand, CS, max F , and total profit. Confirm profit $< \$20$.
2. In the gym example above (Type H: $P_H = 20 - Q_H$; Type L: $P_L = 12 - Q_L$; $MC = \$2$): if the gym cannot observe types and must set a uniform F , which option (B or C) is more profitable? What happens to efficiency under each option?

3. True or false: “Under two-part pricing with identical consumers, both total surplus and consumer surplus are zero.” Explain.

Two-part pricing resolves the DWL problem for a standard monopolist. But what if the monopoly itself is unavoidable due to the cost structure of the industry?

9.4. Natural Monopolies

An electricity grid has a fixed cost of \$10 billion to build and a marginal cost of \$0.05 per kilowatt-hour. Average cost is always declining. If the regulator sets $P = MC$, the grid loses money on every unit sold. What should the regulator do?

9.4.1. Definition and Cost Structure

Why does a natural monopoly arise in the first place? The answer lies entirely in the cost structure. Industries like electricity transmission, water distribution, broadband networks, railway tracks, and gas pipelines share a defining feature: they require enormous fixed investment in physical infrastructure — a national grid, a water pipe network, a fibre-optic backbone — but once that infrastructure is in place, the marginal cost of serving one more customer is very small. A second electricity grid would duplicate every pylon, cable, and substation at enormous social cost, serving no more customers than one grid could. It is cheaper — far cheaper — for society to have a single firm operate the infrastructure than to have two firms each building their own.

This is not a failure of markets in the usual sense. It is a consequence of the technology: fixed costs are so large relative to marginal costs that average cost falls across the entire relevant output range. A single firm is structurally more efficient. The problem is what to do with that monopolist once it exists.

Natural Monopoly: A market where average cost is declining over the *entire* range of relevant output. A single firm can supply the market at lower total cost than two or more firms.

Cost structure: Large fixed costs FC + low marginal cost c :

$$TC = FC + cQ \quad \Rightarrow \quad AC = \frac{FC}{Q} + c > c = MC \text{ for all } Q > 0.$$

AC is always declining; $MC < AC$ everywhere.

In words: The more the firm produces, the more the fixed cost is spread over units, so average cost always falls. But this means any price equal to MC is below average cost — the firm cannot survive.

9.4.2. The Four Regulatory Regimes

Classic examples: electricity grids, water pipes, broadband networks, railway tracks, gas pipelines. High upfront infrastructure, near-zero marginal cost per additional user.

Natural Monopoly: Regulatory Options Compared

Regime	Rule	Q	P	Firm profit	Verdict
Unregulated	$MR = MC$	Q_A (low)	P_A (high)	+ positive	Inefficient; viable
MC pricing	$P = MC$	Q_R (high)	$P_R = MC$	– loss	Efficient; not viable
AC pricing	$P = AC$	Q_{AC}	break-even	0 zero	Nearly efficient; viable
Two-part	$p = MC, F = CS$	Q_{PC}	$MC + \text{fee}$	+ positive	Efficient; viable; $CS = 0$

The regulator's core dilemma: MC pricing is most efficient but requires a subsidy. AC pricing is self-financing but slightly inefficient. Two-part pricing is both efficient and self-financing but extracts all consumer surplus.

Walking through each regime in words.

Regime 1 — Unregulated ($MR = MC$). Left to its own devices, the natural monopolist behaves like any monopolist: it produces where $MR = MC$, charges the profit-maximising price $P_A > AC$, and restricts output. The firm is profitable, but output is well below the efficient level and DWL is large. Consumers who would have been willing to pay more than MC — but not as much as P_A — are excluded from the market entirely. Society loses the value of those trades. This is the regulatory motivation: a legally protected monopoly over essential infrastructure creates sustained, large deadweight losses.

Regime 2 — $P = MC$ (Efficient but loss-making). The regulator mandates that the firm charges price equal to marginal cost. This eliminates all DWL: every consumer willing to pay at least the cost of production can buy. Total surplus is maximised. But since $MC < AC$ everywhere for a natural monopoly, the firm receives revenue per unit that is below average cost per unit. It loses money on every unit sold. Without a government subsidy — funded by taxpayers — the firm exits and the infrastructure collapses. This is the purest efficiency solution, but it is financially unsustainable and politically contentious: taxing the general public to subsidise the electricity company requires justified public finance reasoning and is resisted by fiscal conservatives and free-market advocates alike.

Regime 3 — $P = AC$ (Average-cost/Ramsey pricing: the practical compromise). The regulator sets the price at the level where the firm exactly breaks even: $P = AC$. This is higher than MC , so there is still a small DWL (some consumers who value the good between MC and AC do not buy), but the DWL is much smaller than under unregulated monopoly. Crucially, the firm survives without any subsidy. This is the most common regulatory approach in practice. US state public utility commissions use it for electricity and water; UK regulators like Ofgem and Ofwat use variants of it. The residual inefficiency is the price society pays for financial viability without a public subsidy.

Regime 4 — Two-part pricing ($p = MC, F = CS$). This is the theoretically ideal solution: set the usage charge at MC (achieving efficient output) and charge a fixed connection fee equal to each consumer's surplus. The firm earns profit equal to the entire consumer surplus triangle, making it financially viable with zero DWL. But there are two serious objections. First, equity: the fixed connection fee hits every household equally regardless

In plain English:
 MC pricing = best
 TS, firm needs
 subsidy. AC
 pricing = second
 best, firm
 survives alone.
 Two-part =
 efficient but
 consumers get
 nothing.

of income. A low-income household and a high-income household pay the same access fee, which is regressive. Second, practicality: computing the correct fee requires knowing each consumer's demand, which the regulator typically does not observe. In practice, two-part pricing is used for utilities (monthly meter rental + per-unit charge) as a partial implementation, but the fixed fee is usually set below full CS for equity reasons.

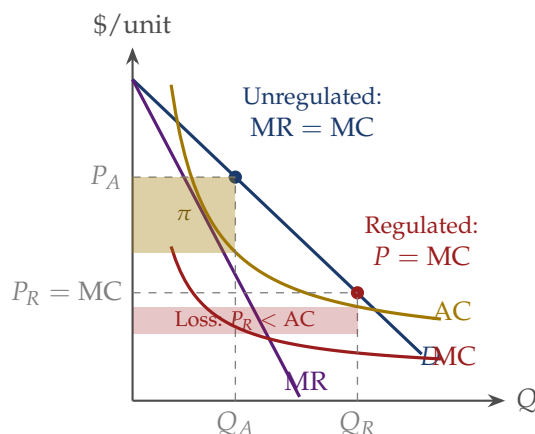


Figure 9.2. Natural monopoly. What to look for: Both AC and MC decline throughout. Unregulated monopolist produces Q_A (low, profitable). $P = MC$ regulation achieves efficient output Q_R but since $P_R < AC(Q_R)$, the firm incurs a loss (red rectangle) requiring a government subsidy.

9.4.3. Cross-Subsidisation

When the regulator permits a natural monopoly to charge different prices to different groups, profits from the high-price group can cover losses on the low-price group — without a government subsidy.

Cross-Subsidisation: Using profits earned from one consumer group (charged above AC) to offset losses from another group (charged near MC). The firm breaks even in aggregate.

In words: Business travellers pay above average cost; leisure travellers pay near marginal cost. Business profits subsidise leisure access. Total firm profit = zero.

Application: Railways and Cross-Subsidisation.

A railway has $MC = \$20$ per passenger and $FC = \$4,000$.

Business travellers (less elastic demand): monopoly price $P_B = \$110$, $Q_B = 45$. Variable profit $= (110 - 20)(45) = \$4,050$.

Leisure travellers (more elastic, regulated at $P_L = MC = \$20$): $Q_L = 80$. Variable profit $= 0$.

Cross-subsidy: business profit (\$4,050) covers fixed costs (\$4,000), leaving a small net surplus. Leisure passengers are served at near-marginal cost without taxpayer funding.

This is third-degree price discrimination applied in a regulatory context: the regulator *requires* above-cost pricing for one group precisely so the firm can afford near-MC pricing for another. The business traveller effectively pays for the rail network's fixed infrastructure.

Why This Matters. Natural monopoly regulation is live policy in electricity, water, broadband, and rail in virtually every country. In the US, state public utility commissions regulate power and water utilities; in the UK, Ofwat regulates water companies. The tension between efficiency and financial viability is not resolved by theory alone — it requires political and institutional trade-offs. Understanding the economics lets you evaluate claims by both regulators and regulated firms about pricing fairness.

Application: How Regulators Handle the Natural Monopoly Dilemma in Practice.

US Electricity Markets. Vertically integrated utilities (generation + transmission + distribution) were historically regulated as natural monopolies by state public utility commissions (PUCs) using *rate-of-return regulation* — a variant of AC pricing that allows the firm to recover all costs plus a “fair” return on capital. Starting in the 1990s, many states deregulated generation (where competition is possible) while keeping transmission and distribution regulated. California’s botched 2000–01 deregulation, which led to rolling blackouts and the Enron energy crisis, is a textbook case of what happens when a network industry is deregulated too fast without adequate market design.

UK Water and Rail. Ofwat (the water regulator) and the Office of Rail and Road regulate private monopolies through *price caps*: the firm is allowed to charge at most a price cap that is updated periodically based on estimated efficient costs. This incentivises cost reduction (if the firm can beat the cap, it keeps the savings) while protecting consumers. UK rail franchising — where private operators pay for the right to run a network they do not own — is a different model: the government retains the infrastructure (a pure natural monopoly) and auctions operating rights competitively.

European Broadband. European regulators require incumbent telecoms firms (BT in the UK, Deutsche Telekom in Germany) to provide access to their physical infrastructure (ducts, cables) at regulated wholesale prices, allowing competitors to offer retail broadband without building their own last-mile infrastructure. This “open access” model tries to harness competition at the retail level while acknowledging the natural monopoly at the infrastructure level.

The common thread: Regulators worldwide face the same trade-off this section describes. None of the four regimes is perfect. The practical choice — AC pricing, price caps, or hybrid models — depends on the specific industry, the information available to regulators, and political judgements about equity and public finance.

Check Your Understanding

1. $P = 60 - Q$, $TC = 200 + 5Q$. Find: unregulated monopoly Q_A , P_A , and profit; regulated Q_R , P_R , and profit or loss.
2. Which of the four regulatory regimes maximises total social welfare? Which is most commonly used in practice, and why is it preferred despite being less efficient?
3. A broadband provider has $TC = 40 + 2Q$ and faces $P = 20 - Q$. Under $P = AC$ regulation, what price does the provider charge and what quantity does it serve? (Hint: set $P = AC$ to get $20 - Q = 40/Q + 2$; rearrange to $Q^2 - 18Q + 40 = 0$.)

We have now seen how a single firm with market power can design pricing. But what happens when

there are two or more strategic firms? That requires game theory.

9.5. An Introduction to Game Theory

Two petrol stations both prefer charging \$2.10/litre. Each knows that if it undercuts to \$1.90 while the other holds, it steals all customers. Both end up at \$1.90 anyway. Why?

9.5.1. Why Standard Optimisation Is Not Enough

In competitive markets and monopoly, each agent maximises their objective taking prices as given. This breaks down in *oligopoly* — a market with a few dominant firms — where each firm's profit depends directly on rivals' choices. **Game theory** provides the framework.

Game Theory: Models strategic behaviour by rational agents who recognise that their payoffs depend on the actions of others.

Every game is defined by three elements:

- (1) **Players:** The decision-making agents.
- (2) **Strategies:** The complete set of actions available to each player.
- (3) **Payoffs:** The outcome each player receives for every combination of strategies.

9.5.2. The Prisoner's Dilemma

Both suspects prefer mutual silence (2 years each). But if one suspects the other will confess, staying silent is catastrophic (10 years). What does each rational player do?

Setup. Two suspects in separate rooms. The DA offers each the same deal:

		Player B	
		Confess	Silent
Player A	Confess	−3, −3	−1, −10
	Silent	−10, −1	−2, −2

In words: (Confess, Confess) = mutual 3 years. (Silent, Silent) = mutual 2 years. Both prefer silence — but each individually prefers to confess regardless of what the other does.

9.5.3. Nash Equilibrium and Dominant Strategies

Nash Equilibrium: A strategy profile (s_A^*, s_B^*) such that each player's strategy is a **best response** to the other's:

$$\pi_A(s_A^*, s_B^*) \geq \pi_A(s_A, s_B^*) \quad \text{for all } s_A$$

Neither player can improve their payoff by deviating *unilaterally*. The equilibrium is **self-enforcing**.

Dominant Strategy: A strategy that is a best response to *every* possible strategy of the opponent. When a dominant strategy exists, a rational player always plays it.

In words: *Nash*: no one wants to deviate given what others are doing. *Dominant*: no one wants to deviate regardless of what others are doing.

All dominant strategy equilibria are Nash equilibria. Not all Nash equilibria involve dominant strategies.

9.5.4. Finding Nash Equilibria: The Circle Method

The Circle Method (Best-Response Analysis):

Step 1. For each *column* (B's strategy), **underline** A's highest payoff in that column.

Step 2. For each *row* (A's strategy), **underline** B's highest payoff in that row.

Step 3. A Nash equilibrium is any cell where **both payoffs are underlined simultaneously**.

Prisoner's Dilemma — Circle Method.

		Confess	Silent
A	Confess	<u>-3</u> , <u>-3</u>	<u>-1</u> , -10
	Silent	-10, <u>-1</u>	-2, -2

A's best responses: if B Confesses $\rightarrow$ Confess ($-3 > -10$); if B Silent $\rightarrow$ Confess ($-1 > -2$). **Confess is dominant for A.**

B's best responses: symmetric. **Confess is dominant for B.**

Both payoffs underlined at (Confess, Confess) = $(-3, -3)$. **Nash equilibrium: (Confess, Confess).**

Note: (Silent, Silent) = $(-2, -2)$ would be better for both — but it is not a Nash equilibrium, since each player could improve by deviating to Confess.

The Tragedy of the Prisoner's Dilemma. Individual rationality drives both players to (Confess, Confess) = $(-3, -3)$, which is **Pareto dominated** by (Silent, Silent) = $(-2, -2)$. There exists an outcome that makes *everyone* better off, yet it is unstable because each player has a private incentive to defect. Collective irrationality emerges from individual rationality.

Practice Game: Verifying Nash Equilibrium.

		Left	Right
A	Top	<u>3</u> , <u>9</u>	<u>2</u> , 8
	Bottom	0, 2	1, 1

A: Top dominates ($3 > 0$ vs Left; $9 > 1$ vs Right). B: Left is best response to Top ($9 > 8$); Left is best response to Bottom ($2 > 1$). Left dominates for B. **NE: (Top, Left) with payoffs (3, 9).** Both payoffs underlined. ✓

9.5.5. Real-World Applications

Application: OPEC as a Prisoner's Dilemma.

OPEC members agree to restrict oil output to keep prices high. Each member has a private incentive to overproduce: selling more at the cartel price earns extra profit. If all members

overproduce, the price collapses and everyone is worse off.

Numbers: If both restrict: each earns \$900M/day. If one overproduces while the other restricts: overproducer earns \$1,200M, restrictor earns \$300M. If both overproduce: each earns \$600M.

		Restrict	Overproduce
Country A	Restrict	900, 900	300, 1200
	Overproduce	1200, 300	<u>600, 600</u>

NE: (Overproduce, Overproduce) = (600, 600). Overproduce is dominant for each member ($1200 > 900$ vs Restrict; $600 > 300$ vs Overproduce). Yet (Restrict, Restrict) = (900, 900) Pareto dominates — the cartel outcome is collectively superior but individually unstable. This is why OPEC agreements are perpetually fragile and why enforcement mechanisms (monitoring, credible punishment) are essential to cartel stability.

Check Your Understanding

1. Apply the circle method to the OPEC matrix. Confirm (Overproduce, Overproduce) is the unique NE.
2. In a game where neither player has a dominant strategy, can there still be a Nash equilibrium? Give an example.
3. True or false: “A Nash equilibrium is always the best outcome for both players.”
4. In the prisoner’s dilemma, if both players could communicate and sign a *binding* agreement, what would they choose? Why do non-binding agreements fail to sustain cooperation in a one-shot game?

The prisoner’s dilemma structure appears in pricing wars, advertising battles, arms races, and the tragedy of the commons. But not all strategic games have a unique equilibrium. A second game form — the coordination game — has multiple Nash equilibria and raises a different kind of strategic problem.

9.5.6. Coordination Games: When There Are Multiple Nash Equilibria

Not all games with strategic interdependence look like the prisoner’s dilemma. In the prisoner’s dilemma, the problem is that individual incentives push players toward a uniquely bad outcome. In a **coordination game**, the problem is different: there are *multiple Nash equilibria*, and the challenge is that players must coordinate on the *same* equilibrium without being able to communicate. Individual rationality does not fail in the usual sense — any of the Nash equilibria is stable once reached — but there is no guarantee that players will end up at the Pareto-superior one.

Coordination Game: A game in which there are multiple Nash equilibria and each player’s best response depends on what the other player chooses. Players prefer to coordinate on the same equilibrium but may fail to do so without communication.

In words: In a prisoner’s dilemma: both are drawn to a unique bad equilibrium. In a coordination game: both want to match each other, but there are multiple places they could match.

Technology Standard Adoption.

Two firms (Apple and Samsung) must each choose whether to adopt Standard A or Standard B for a new device protocol. Compatibility is valuable: if both adopt the same standard, consumers benefit and both firms profit. If they adopt different standards, neither network forms and both lose.

		Left	Right
A	Up	8, 8	2, 3
	Down	3, 2	<u>6</u> , <u>6</u>

(Row = Apple. Strategies: Up = Standard A, Down = Standard B; Left = Standard A, Right = Standard B.)

Best-response analysis:

- Apple: if Samsung plays Standard A (Left), Apple prefers Standard A ($8 > 3$). If Samsung plays Standard B (Right), Apple prefers Standard B ($6 > 2$). No dominant strategy.
- Samsung: symmetric. No dominant strategy.

Nash equilibria: Two — **(Standard A, Standard A)** = (8, 8) and **(Standard B, Standard B)** = (6, 6). Both are self-enforcing: once both firms are at the same standard, neither wants to deviate unilaterally (they would lose the network benefit).

Key insight: (Standard A, Standard A) = (8, 8) **Pareto dominates** (Standard B, Standard B) = (6, 6) — everyone prefers Standard A. But if Samsung believes Apple will adopt Standard B, Samsung's best response is Standard B too. The Pareto-superior equilibrium is not guaranteed.

Contrast with prisoner's dilemma:

- Prisoner's dilemma: ONE Nash equilibrium, but it is Pareto dominated by the cooperative outcome (which is *not* a NE).
- Coordination game: TWO Nash equilibria; the problem is selecting the right one, not escaping from a bad one.

Coordination Failure. When the Pareto-superior NE is not reached, we call it a **coordination failure**. Historical examples abound: VHS vs. Betamax, Blu-ray vs. HD DVD, QWERTY vs. Dvorak keyboards, AC vs. DC electrical current (Tesla vs. Edison). In each case, the technology standard adopted may not have been the technically superior one — but once one standard achieved critical mass, the network effect locked it in. From a business strategy perspective, being first to achieve scale ("first mover advantage") can determine which equilibrium the market coordinates on.

Check Your Understanding

1. In the technology standard game above, apply the circle method to confirm the two Nash equilibria. Is there a dominant strategy for either player?
2. Name one real-world market where coordination failure led to an inefficient

Coordination games apply beyond technology: two firms deciding whether to enter the same market, two countries choosing whether to adopt the same trade standard, two departments in a firm choosing compatible software.

standard being locked in. What would have been required to reach the Pareto-superior outcome?

3. Distinguish: (a) a prisoner's dilemma from (b) a coordination game. For each, state the number of Nash equilibria and whether the NE is Pareto optimal.

The prisoner's dilemma structure appears in pricing wars, advertising battles, arms races, and the tragedy of the commons. We review these before turning to the practice questions.

9.5.7. Further Applications of the Prisoner's Dilemma

- **Price wars:** (Low, Low) is the NE of a one-shot pricing game. Both firms prefer (High, High), but each has an incentive to undercut. This is why price competition drives prices toward marginal cost.
- **Advertising wars:** Both advertising is the NE, but mutual restraint saves cost while maintaining market share. Each firm has an incentive to advertise more than is collectively optimal.
- **Arms races:** Mutual restraint produces the best collective outcome; each nation has an incentive to arm unilaterally, driving both to heavy military spending.
- **The commons problem:** Each fisherman harvests as much as possible before others do. The NE involves excessive harvesting and eventual resource depletion.

With all content covered, we close with the exam materials.

Where Does This Take Us?

- **Chapter 10 — Oligopoly Models:** The game-theoretic tools introduced here — Nash equilibrium, best-response analysis, dominant strategies — are the foundation of Chapter 10's oligopoly models. Cournot competition (firms choose quantities simultaneously) and Bertrand competition (firms choose prices simultaneously) are both formally games where firms are the players and quantities or prices are the strategies. The Nash equilibrium in the Cournot model is the outcome of each firm playing its best-response quantity function; the result is a prisoner's dilemma where both firms would prefer to restrict output but each has an incentive to expand.
- **Coordination games and standards:** Chapter 10 will also revisit coordination failures in the context of market entry and spatial competition, where firms must decide where to locate knowing that location choices interact.
- **Repeated games:** When the same firms interact repeatedly, cooperation (High, High) can be sustained by "grim trigger" strategies — cooperate until the other deviates, then punish forever. The logic requires that the present value of continued cooperation exceeds the one-time gain from defecting.
- **Entry deterrence:** An incumbent's threat to fight entry is credible only if fighting is a best response after entry occurs. Sequential game theory (sub-game perfect equilibrium) disciplines which threats are credible — the tools are introduced in the practice questions;

the full application is in Chapter 10.

- **Two-part pricing in practice:** The doubled-profit result ($\pi_{2P} = 2\pi_M$) holds exactly for linear demand. For non-linear demand the result is approximate but directionally robust; see the Level 3 questions.

Seven Common Exam Mistakes — and How to Avoid Them

1. **Setting $p = 0$ (or $p < MC$) in two-part pricing.** Lower usage charges do not always mean more profit. At $p < MC$, the firm loses money on every unit sold. The loss on units more than offsets any increase in the fixed fee. Always set $p^* = MC$ — not $p^* = 0$.
2. **Confusing F^* (the optimal fixed fee) with profit.** $F^* = CS(p = MC)$ is the *maximum* fixed fee consumers will pay — it equals their consumer surplus at the efficient price. The firm's *profit* from two-part pricing is $F^* + (p^* - MC) \cdot Q = F^*$ (since $p^* = MC$, the per-unit margin is zero). Do not confuse the fixed fee with the profit rectangle from monopoly pricing: $F^* = CS_{PC}$, not π_M .
3. **Forgetting the natural monopoly loss at $P = MC$.** Because $MC < AC$ for all output levels in a natural monopoly, setting $P = MC$ always means $P < AC$ and therefore a per-unit loss. A natural monopoly cannot survive at marginal-cost pricing without a subsidy — *always* check whether P covers AC .
4. **Equating Nash equilibrium with the best outcome.** The prisoner's dilemma shows that the Nash equilibrium (Confess, Confess) = $(-3, -3)$ is Pareto dominated by (Silent, Silent) = $(-2, -2)$. Nash equilibrium is the *stable* outcome — not the *best* outcome. Stability and optimality are distinct concepts.
5. **Applying dominant-strategy logic to games without dominant strategies.** Not every game has a dominant strategy equilibrium. In a coordination game, the best action for each player depends on what the other player does. Use the circle method systematically — do not assume dominant strategies exist before checking.
6. **Confusing coordination games with prisoner's dilemmas.** A prisoner's dilemma has ONE Nash equilibrium (which is Pareto suboptimal). A coordination game has MULTIPLE Nash equilibria (at least one of which may be Pareto dominant). The strategic problem is different: in a PD, the issue is escaping the bad NE; in a coordination game, the issue is selecting the good NE.
7. **Claiming AC pricing is allocatively efficient.** Average-cost pricing ($P = AC$) is *nearly* efficient — it produces more output than the unregulated monopolist — but it is still below the efficient level Q_R (where $P = MC$). Only $P = MC$ regulation (or two-part pricing) achieves allocative efficiency. AC pricing involves a small residual DWL.

Chapter at a Glance — Week 9: Pricing, Natural Monopoly & Game Theory

Your one-page revision guide.

Key Equations & Results

$T(q) = F + pq$	Two-part pricing
$p^* = MC$	Optimal usage charge
$F^* = CS(p = MC)$	Optimal fixed fee
$\pi_{2P} = 2\pi_M$	Profit doubles
$TC = F + cQ \Rightarrow AC > c = MC$	Natural monopoly
$\pi_A(s_A^*, s_B^*) \geq \pi_A(s, s_B^*)$	Nash equilibrium

Three-Scenario Table (benchmark $P = 3 - Q/10$, $MC = 1$)

Scenario	Q	CS	π	DWL
Monopoly	10	\$5	\$10	\$5
Competition	20	\$20	\$0	\$0
Two-part	20	\$0	\$20	\$0

Natural Monopoly Regimes

Regime	Efficient?	Viable?	CS
Unregulated	No	Yes	Low
$P = MC$	Yes	No (loss)	High
$P = AC$	Nearly	Yes	Medium
Two-part	Yes	Yes	Zero

Prisoner's Dilemma

		Confess	Silent
A	C	-3, -3	-1, -10
	S	-10, -1	-2, -2

NE: (C,C) = (-3, -3). Pareto dominated by (S,S) = (-2, -2).
Dominant strategy: Confess.

Circle Method Summary

Underline A's best payoff in each column.
Underline B's best payoff in each row.
Cell with *both* underlined = Nash equilibrium.

Coordination Game

Multiple NE: e.g. (A,A) = (8, 8) and (B,B) = (6, 6).

No dominant strategy. Pareto-superior NE not guaranteed.

Coordination failure: players may land at inferior NE.

LO Checklist

- LO1: Linear vs. non-linear; $T(q) = F + pq$ ✓
- LO2: Two-step rule: $p^* = MC$, $F^* = CS$ ✓
- LO3: $\pi_{2P} = 2\pi_M$ for linear demand ✓
- LO4: Four regulatory regimes; trade-offs ✓
- LO5: Cross-subsidisation logic ✓
- LO6: NE, dominant strategy, circle method ✓
- LO7: Prisoner's dilemma structure ✓
- LO8: Coordination game; multiple NE; coordination failure ✓

The thread: Uniform monopoly (Week 8) leaves CS uncaptured $\rightarrow$ two-part pricing ($p = MC$, $F = CS$) captures all CS, doubles profit, eliminates DWL $\rightarrow$ natural monopoly: $AC > MC$ always, so $P = MC$ means losses $\rightarrow$ AC pricing breaks even but is slightly inefficient; two-part pricing is efficient and profitable but $CS = 0 \rightarrow$ game theory: with multiple strategic firms, optimisation is insufficient; best-response analysis finds Nash equilibrium $\rightarrow$ prisoner's dilemma: individual rationality $\neq$ collective optimality.

9.6. Key Terms

Non-Linear Pricing

Pricing where the per-unit price depends on quantity; $T(q)/q$ is not constant. *Used throughout the course.*

Two-Part (Two-Tier) Pricing

Fixed fee F plus per-unit charge p : $T(q) = F + pq$.

Fixed Fee (Entry Fee)

Lump-sum charge to gain access, independent of quantity consumed.

Usage Charge (Per-Unit Price)

Variable charge p per unit; should equal MC at the optimum.

Optimal Two-Part Rule

$p^* = MC$; $F^* = CS(p = MC)$; profit = $2\pi_M$ for linear demand.

Natural Monopoly

Market where AC declines throughout; arises from large fixed costs relative to marginal costs.

Marginal-Cost Regulation

$P = MC$: efficient but forces a loss since $MC < AC$.

Average-Cost Regulation (Ramsey Pricing)

$P = AC$: firm breaks even; nearly efficient (small residual DWL). The practical compromise between allocative efficiency and financial viability for a regulated natural monopoly.

Cross-Subsidisation

Profits from one consumer group (above AC) offset losses from another (near MC); firm survives without a government subsidy.

Game Theory

Models strategic interactions among rational agents whose payoffs depend on others' choices.

Players

Decision-making agents in a game.

Strategies

The complete set of actions available to each player.

Payoffs

Outcomes each player receives for every strategy combination.

Payoff Matrix

Tabular representation: rows = A's strategies, columns = B's strategies, cells = pay-off pairs (A, B).

Best Response

The strategy maximising a player's payoff given what the other player does.

Dominant Strategy

Best response to *every* opponent strategy; always play it when it exists.

Nash Equilibrium

Strategy profile where each player's strategy is a best response to the other's; no unilateral deviation is profitable. *Used throughout the course.*

Prisoner's Dilemma

Game where dominant strategies lead to a Nash equilibrium Pareto dominated by the cooperative outcome; illustrates conflict between individual and collective rationality.

Coordination Game

A game with multiple Nash equilibria in which each player's best response depends on the other's strategy; players prefer to coordinate on the same equilibrium but may fail without communication.

Coordination Failure

Outcome where players end up at a Pareto-inferior Nash equilibrium because they cannot coordinate on the Pareto-superior one. Classic examples: technology standard wars, market entry games.

Pareto Dominance

Outcome X Pareto dominates Y if every player is at least as well off under X, and at least one is strictly better off.

Credible Threat

A threat the threatening player would actually carry out; relevant for entry deterrence and repeated games.

Recommended Videos

Four videos selected for clarity and direct relevance to this chapter.

1. Two-Part Tariffs (Matt Birch) ~ 10 min

<https://www.youtube.com/watch?v=eZMJbHA1fBo>

Derives the optimal two-part pricing rule from scratch, explains why $p^* = MC$ maximises fixed-fee revenue, and compares outcomes across pricing regimes. Reinforces Section 9.3 and is recommended before Level 2 two-part questions.

2. Natural Monopoly and Government Regulation (Jason Welker) ~ 9 min

<https://www.youtube.com/watch?v=LhiyJlI5xa0>

Covers declining average cost, the loss at $P = MC$, AC pricing, and the regulatory dilemma. Directly reinforces Section 9.4 and is recommended before Level 2 natural monopoly questions.

3. Prisoners' Dilemma and Nash Equilibrium (Khan Academy) ~ 11 min

<https://www.youtube.com/watch?v=UkXI-zPcDIM>

Derives Nash equilibrium via best-response analysis, uses the prisoner's dilemma as the central example, and explains why cooperative outcomes are unstable. Reinforces Section 9.5 and recommended before Level 2 game theory questions.

4. Game Theory of Cheating Firms (Khan Academy) ~ 8 min

<https://www.youtube.com/watch?v=XTMbRDRaviM>

Applies the prisoner's dilemma framework to cartel pricing and shows how the incentive to cheat generates collectively suboptimal industry equilibria. Reinforces Section 9.5 and the applications in the chapter.

9.7. Practice Questions

How to use these questions. Difficulty: ★ easy ★★ intermediate ★★★ challenging. Payoffs written as (Row player, Column player); higher is better unless stated. Attempt every question independently before consulting solutions.

Level 1: Conceptual and Short-Answer

★ [Linear vs. Non-Linear Pricing]

- (a) Define linear pricing. Give one everyday example.
- (b) Define non-linear pricing. Why does a firm with market power prefer it to a uniform price?
- (c) A gym charges \$50/month plus \$5/class. Linear or non-linear? Identify F and p .
- (d) 500 g flour costs \$2.00; 1 kg costs \$3.60. Compute the per-gram price in each case. Is this linear pricing?

★ [Two-Part Pricing — Concepts]

- (a) Describe the two components of two-part pricing. In what sense is it price discrimination?
- (b) Why does the optimal two-part pricing rule set the per-unit charge equal to MC?
- (c) Under optimal two-part pricing, $CS = 0$. Does this mean the scheme is allocatively inefficient?
- (d) Give three real-world examples. For each, identify F and p .

★ [Natural Monopoly — Concepts]

- (a) Define a natural monopoly. What cost feature distinguishes it from a standard monopoly?
- (b) Why does marginal-cost regulation force a natural monopolist to incur losses? Use the $MC < AC$ relationship.
- (c) Compare unregulated monopoly vs. $P = MC$ with subsidy from the perspectives of efficiency, consumer welfare, and public finance.
- (d) How does cross-subsidisation allow a regulated natural monopoly to survive without a subsidy?

★ [Game Theory — Concepts]

- (a) What are the three elements that define a game? Briefly describe each.
- (b) What is a Nash equilibrium? In what sense is it “self-enforcing”?
- (c) What is a dominant strategy? If both players have dominant strategies, is their

profile always a Nash equilibrium?

- (d) Why do rational players end up at $(\text{Confess}, \text{Confess}) = (-3, -3)$ when $(\text{Silent}, \text{Silent}) = (-2, -2)$ is better for both?

★ **[Prisoner's Dilemma in Business]** Two petrol stations can charge \$2.00/L (High) or \$1.60/L (Low). Both High: each earns \$800/day. One undercuts: cutter earns \$1,100, holder earns \$200. Both Low: each earns \$500.

- (a) Construct the payoff matrix.
- (b) Find the Nash equilibrium. Is this a prisoner's dilemma?
- (c) Does either player have a dominant strategy?
- (d) Why is the cooperative (High, High) agreement difficult to sustain?

★ **[Two-Part Pricing: Equity]**

- (a) A consumer group argues two-part pricing harms low-income households even though it is allocatively efficient. Explain using CS and the fixed fee.
- (b) Compare distributional outcomes of (i) standard monopoly, (ii) two-part pricing, (iii) $P = MC$ with subsidy. Which leaves the most CS? Which is best for the firm?
- (c) A regulator caps the fixed fee at half the full CS. How does this affect profit and CS relative to the uncapped optimum?

★ **[Nash Equilibrium and Strategic Thinking]**

- (a) Can a game without dominant strategies still have a Nash equilibrium? If yes, how do you find it?
- (b) Two firms simultaneously decide whether to enter a new market. If both enter: each earns $-\$10$. Only one enters: it earns $\$50$; the other earns $\$0$. Neither enters: both earn $\$0$. Construct the payoff matrix and find all Nash equilibria.
- (c) Is the game in (b) a prisoner's dilemma, coordination game, or neither?
- (d) Why does cooperation sometimes emerge in repeated games but not one-shot games?

★ **[Coordination Games — Concepts]**

- (a) Define a coordination game. How does it differ from a prisoner's dilemma in terms of (i) number of Nash equilibria and (ii) whether the NE is Pareto optimal?
- (b) In a coordination game with two Nash equilibria, both of which are pure-strategy NE, how does the concept of Pareto dominance help identify which equilibrium is "better"? Does Pareto dominance guarantee that players reach the superior equilibrium?

- (c) Name two real-world examples of coordination games (other than the technology standard example in the text). For each, identify the players, strategies, and why coordination failure is a risk.
- (d) A firm is considering adopting a new enterprise software platform. It is more productive than the current system, but only if its key suppliers also adopt it. What type of strategic situation is this? What could the firm do to increase the probability of reaching the efficient outcome?

★ **[Explain to a Friend — Level 1]** *No formulas.* Your friend asks: “If two-part pricing is efficient (no DWL) and earns the firm more profit, why would anyone ever complain about it?” Write 3–4 plain-English sentences explaining the distributional concern.

Level 2: Intermediate Problems

★★ **[Two-Part Pricing: Basic]** $P = 10 - Q$, $MC = AC = \$2$.

- (a) Standard monopoly: Q_M^* , P_M^* , π_M .
- (b) PC: Q_{PC} , P_{PC} , CS_{PC} .
- (c) Optimal two-part: F^* , p^* , total profit.
- (d) Verify $\pi_{2P} = 2\pi_M$. Explain why.
- (e) Show two-part pricing eliminates DWL.

★★ **[Two-Part Pricing: Welfare Table]** $P = 20 - 2Q$, $MC = AC = \$4$.

- (a) Standard monopoly: Q_M^* , P_M^* , π_M , CS_M , DWL_M .
- (b) PC: Q_{PC} , CS_{PC} .
- (c) Two-part: F^* , p^* , total profit.
- (d) Fill in the three-scenario table (Q , CS , π , TS , DWL).
- (e) Explain why two-part pricing achieves the same TS as PC but gives it entirely to the monopolist.

★★ **[Two-Part Pricing: General Linear Demand]** $P = a - bQ$, $MC = c$.

- (a) Standard monopoly: Q_M^* , P_M^* , π_M .
- (b) PC: Q_{PC} , CS_{PC} .
- (c) Two-part: F^* , p^* , π_{2P} .
- (d) Prove $\pi_{2P} = 2\pi_M$.

★★ **[Two-Part Pricing: Cover Charge]** A nightclub consumer has demand $Q = 8 - P$. Drinks cost $MC = AC = \$2$.

- (a) Drinks bought at $p = \$2$?
- (b) CS at $p = \$2$?
- (c) Optimal cover charge F^* and per-drink price p^* .
- (d) Total profit per consumer per evening.
- (e) Uniform monopoly price? Compare profit and CS.

★★ **[Data-Based: Two-Part Pricing vs. Monopoly]** The table shows a monopolist's demand data ($MC = AC = \$4$, constant).

Q	P	TR	MR	Profit (π)
0	20	0	—	—
2	18	36		
4	16	64		
6	14	84		
8	12	96		
10	10	100		
12	8	96		

- (a) Fill in MR (per-unit, between consecutive rows) and profit ($\pi = TR - TC$, with $TC = 4Q$).
- (b) Find the profit-maximising output. Compute Q_M^* , P_M^* , π_M .
- (c) Identify the competitive output where $P = MC = \$4$. Calculate CS_{PC} .
- (d) Under two-part pricing: find F^* , p^* , and total profit. Confirm profit $= 2\pi_M$.
- (e) Compute DWL under standard monopoly. Confirm it is zero under two-part pricing.
- (f) At which output is TR maximised? What is MR there? What does this imply about elasticity?

★★ **[Natural Monopoly: Basic]** $P = 60 - Q$, $TC = 200 + 5Q$.

- (a) Unregulated monopoly: (Q_A, P_A) and profit.
- (b) $P = MC$ regulation: Q_R , P_R , and profit (or loss).
- (c) $P = AC$ break-even output.
- (d) Comparison table: Q , P , profit, CS, TS for all three regimes.
- (e) Which is allocatively efficient? Which is financially sustainable without subsidy?

★★ **[Game Theory: Finding Nash Equilibria]** For each payoff matrix, find all Nash equi-

libria and dominant strategies.

		L		R
(a)	A	Up	4, 3	2, 1
		Down	1, 2	3, 4

		L		R
(b)	A	Up	5, 5	0, 8
		Down	8, 0	3, 3

		L		C	R
(c)	A	Up	2, 1	4, 3	3, 2
		Down	1, 4	3, 2	2, 3

★★ [Two-Part Pricing: Heterogeneous Consumers] A gym is the only fitness facility in town. Half the members are **Fitness Nuts** (Type H) with individual demand $P_H = 20 - Q_H$. Half are **Casual Exercisers** (Type L) with individual demand $P_L = 12 - Q_L$. $MC = AC = \$2$ per visit. Types are not directly observable.

- Compute CS_H and CS_L at $p = MC = \$2$.
- Option A (type-specific fees, if observable):** Set $F_H = CS_H$ and $F_L = CS_L$, $p = \$2$ for both. Compute per-pair profit.
- Option B (uniform fee, retain both types):** Set the largest F that keeps both types in the market. Compute per-pair profit. What is the *information rent* earned by Type H consumers?
- Option C (exclude low types):** Set $F = CS_H$; only Fitness Nuts join. Compute per-pair profit.
- Which option maximises per-pair profit? Interpret: what does this say about the value of consumer information to the firm?
- Under Option B, is the market allocation efficient? What happens to DWL?

★★ [Coordination Game: Technology Standards] Two tech firms (Apex and Bolt) must each independently choose to adopt **Standard X** or **Standard Y** for a new wireless protocol. The payoff matrix (Apex's payoff, Bolt's payoff) is:

		Standard X	Standard Y
Apex	Standard X	10, 10	1, 3
	Standard Y	3, 1	7, 7

- Apply the circle method. Find all Nash equilibria.
- Does either player have a dominant strategy?
- Which NE is Pareto dominant? Is it necessarily the one that will be reached?

(d) Apex moves first and publicly announces it will adopt Standard X. How does this commitment affect the game's outcome? What game-theoretic concept does this illustrate?

(e) Is this a prisoner's dilemma or a coordination game? Justify.

★★ **[Oligopoly as Prisoner's Dilemma]** Two firms choose Restrict (10 units) or Expand (20 units). Market demand: $P = 50 - Q_\alpha - Q_\beta$. $MC = AC = \$10$.

(a) Compute price and profit for all four combinations.

(b) Construct the payoff matrix.

(c) Nash equilibrium? Dominant strategies?

(d) Is this a prisoner's dilemma? Show cooperative outcome Pareto dominates.

(e) If firms sign a binding agreement, what restriction maximises joint profit?

★★ **[Explain to a Friend — Level 2]** Your friend says: "I understand the prisoner's dilemma in theory, but I don't see why petrol stations don't just agree to charge the high price and stick to it." Write a 4–5 sentence explanation of why a one-shot agreement is not self-enforcing, using the concepts of dominant strategy and unilateral deviation.

Level 3: Advanced Problems

★★★ **[Two-Part Pricing: Multiple Consumer Types]** A golf club has Type H (demand $Q_H = 12 - p$) and Type L ($Q_L = 6 - p$). $MC = \$2/\text{round}$. Equal numbers of each type.

(a) $CS_H(p = 2)$ and $CS_L(p = 2)$.

(b) Types identifiable: optimal (F_H, p_H) and (F_L, p_L) . Total per-pair profit.

(c) Types unobservable, uniform F : setting $p = \$2$, max F without excluding low-types. Per-pair profit.

(d) Exclude low-types: $F = CS_H$. Per-pair profit.

(e) Which strategy maximises per-pair profit? What is the trade-off?

★★★ **[Natural Monopoly: Full Welfare Analysis]** $P = 100 - 2Q$, $TC = 400 + 4Q$.

(a) Verify AC is declining for all $Q > 0$.

(b) Unregulated monopoly: Q_M^* , P_M^* , π_M , CS_M , TS_M .

(c) $P = MC$: Q_R , P_R , firm profit/loss, CS_R , TS_R .

(d) $P = AC$: Q_{AC} , P_{AC} , CS_{AC} , TS_{AC} .

(e) Required subsidy under $P = MC$? Efficiency cost of AC pricing?

(f) Summarise all three in a table.

★★★ **[Cross-Subsidisation: Railway]** Business travellers: $P_B = 200 - 2Q_B$. Leisure travellers: $P_L = 100 - Q_L$. $MC = \$20$; $FC = \$4,000$.

- Monopoly price and output for business travellers; variable profit.
- Monopoly price and output for leisure travellers; variable profit.
- Total net profit (both groups, net of FC). Profitable?
- Regulator requires $P_L = MC = \$20$. Variable loss on leisure passengers?
- Under (d), what business price allows the company to exactly break even?
- Explain how cross-subsidisation makes the scheme sustainable.

★★★ **[Comparing Pricing Strategies]** Software company: Students ($Q_S = 100 - P_S$), Professionals ($Q_P = 80 - 0.5P_P$). $MC = \$20$.

- Uniform price: find monopoly P^* , Q^* , total profit.
- Market separation: (Q_S^*, P_S^*) , (Q_P^*, P_P^*) , total profit.
- Two-part pricing (aggregate demand): Q_{PC} , CS_{PC} , F^* , total profit.
- Rank profits across the three strategies.
- Under what conditions does two-part pricing dominate market separation?

★★★ **[Full Natural Monopoly: Water Utility]** $P = 80 - Q$, $TC = 320 + 8Q$.

- Confirm natural monopoly (show AC declining).
- Unregulated monopoly: Q_M^* , P_M^* , π_M , CS_M , DWL_M .
- $P = MC$: Q_R , P_R , firm profit, CS_R , TS_R .
- Two-part ($p = MC$, $F = CS$): F^* , total profit. Does the utility break even?
- $P = AC$: Q_{AC} , CS_{AC} , TS_{AC} .
- Comprehensive table: all four regimes (unregulated, $P = MC$, $P = AC$, two-part).
- Recommend a regulatory approach. Justify using efficiency and equity arguments.

★★★ **[Game Theory: R&D and Strategy]** TechX and TechY each decide to Invest or Not: Both invest $\rightarrow (20, 20)$. Only one invests $\rightarrow (50, 10)$ for investor. Neither $\rightarrow (30, 30)$.

- Write the payoff matrix.
- Find best responses and dominant strategies.
- Find all Nash equilibria.

(d) Prisoner's dilemma, coordination game, or neither? Justify.

(e) Both prefer (Don't, Don't). Is it a NE? Explain the tension.

★★★ [Integrated: Pricing, Regulation, and Entry] Streaming service: $P = 50 - 0.5Q$, $MC = AC = \$10$.

(a) Standard monopoly: (Q_M^*, P_M^*) , π_M , CS_M , DWL_M .

(b) Two-part pricing: optimal (F^*, p^*) , total profit.

(c) Entry game (payoffs as below). Find all Nash equilibria. Is the Fight threat credible?

		Enter	Stay Out
Incumbent	Accommodate	20, 15	32, 0
	Fight	0, -5	20, 0

(d) What does "credible threat" mean in entry deterrence? Under what conditions could Fight be credible?

★★★ [Explain to a Friend — Level 3] Your friend, a business student, says: "Natural monopoly regulation is just interference in the market. If the grid loses money at $P = MC$, it should just charge what the market will bear." Write a 5–6 sentence response addressing: (a) why the regulator might mandate $P = MC$; (b) why this creates a financial problem; (c) what AC pricing achieves as a compromise; (d) whether two-part pricing solves the dilemma and at what cost; and (e) what a reasonable regulator should actually do.

Full solutions follow in the solutions document. Attempt all questions independently before consulting them.